Expanding Polynomials & Pascal's Triangle

Algorithm for Expanding Binomials Raised to the *n*th Power

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Churchill College | IGCSE Mathematics

Session Flow: Polynomial Expansion (100 min)

A student-centered sequence based on Buckingham's PGC Framework.

Lesson Architecture

- **10 min – Phase 1:** Novelty & Curiosity (Geometric Puzzle & Exponent Laws)
- **20 min – Phase 2:** Expectation & Motivation (FOIL & Pascal Induction)
- **45 min – Phase 3:** Effort & Scaffolding (Level 1, Level 2, Level 3 Work)
- **25 min – Phase 4:** Recognition & Feedback (Peer Audit & Synthesis)

1. Novelty & Curiosity | The Geometric & Exponent Trap

Driving Question (Think-Pair-Share – 4 min)

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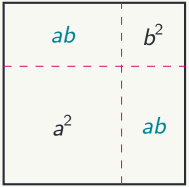

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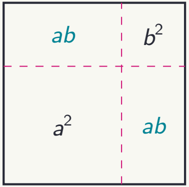

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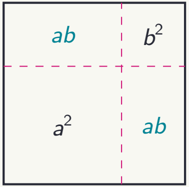

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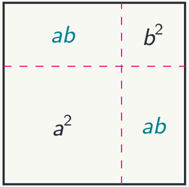

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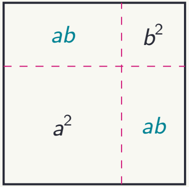

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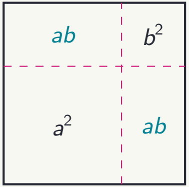

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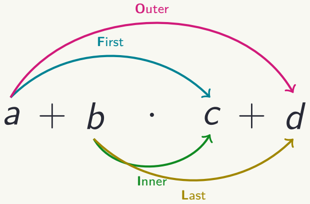
2. Expectation & Motivation | The FOIL Structural Rule

FOIL systematically distributes every term in the first binomial to the second: $(a+b)(c+d)$



General FOIL Formula

$$(a+b)(c+d) = ac + ad + bc + bd$$

When $c = a$ and $d = b$: $\quad (a+b)(a+b) = (a+b)^2 = a^2 + ab + ab + b^2 = \mathbf{a^2} + \mathbf{2ab} + \mathbf{b^2}$

Using the **FOIL** method expand the following expressions:

1. Expand and simplify:

$$(3x + 5)(2x - 7)$$

2. Expand and simplify:

$$(x^2 + 3x - 4)(2x + 5)$$

3. Expand and simplify:

$$(2x^2 - x + 3)(x^2 + 4x - 2)$$

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Guided Discovery (Pair Work – 8 min): Expand and simplify the following expressions in your notebooks.

$$(a+b)^0 =$$
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Pascal's Triangle

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ 1 & & 3 & & 3 & & 1 \end{array}$$

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			1			
		1		1		
	1		2		1	
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Symmetric Variable Rule

Notice: As exponent of a **decreases** $(3 \to 2 \to 1 \to 0)$, exponent of b **increases** $(0 \to 1 \to 2 \to 3)$.

3. Effort & Scaffolding | Binomial Expansion (18 min)

[Level 2: Intermediate – Pair Work]

Use Pascal's Triangle coefficients to expand:

Squares & Cubes:

1. $(x + 3)^2$
2. $(2x^2 - 5y)^2$
3. $(x^3 + 2y^3)^3$
4. $(2xz - 4z)^3$
5. $(x + xy)^3$
6. $(-3x + x^3)^3$

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Algorithm

1. **Step1:** Use the Pascal's Triangle in order to obtain the general formula.
2. **Step 2:** Substitute the corresponding values on the formula.
3. **Step 3:** Simplify by using the laws of the exponents
   3.1 $a^0 = 1$ if $a \neq 0$
   3.2 $a^1 = a$
   3.3 $a^n \cdot a^m = a^{n+m}$
   3.4 $(a^n)^m = a^{nm}$
   3.5 $(a \times b)^n = a^n \times b^n$

3. Effort & Scaffolding | Level 3: Trinomial Challenge (15 min)

[Level 3: Hard / Advanced – Small Groups]

Problem: How do we expand a squared trinomial like $(a+b+c)^2$ without long multiplication?

Grouping Strategy:

- Treat $(a+b)$ as a single block u.
- Expand $(\underbrace{a+b}_{u}+c)^2 = (u+c)^2 = u^2 + 2uc + c^2$.
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$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

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Group Tasks:

1. Expand $(x+2y-z)^2$ in your notebooks.
2. **High-Level Extension:** Predict the structure of $(a+b+c+d)^2$.

4. Recognition & Feedback | Peer Audit Protocol (15 min)

Notebook Swap: Exchange notebooks with another pair. Audit their Level 2 and Level 3 work.

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Solution: $x^2+4y^2+z^2+4xy-2xz-4yz$

1. **Sign Audit:** Check if negative signs were squared correctly ($(-z)^2=+z^2$) and cross-terms preserve signs (there are no problems with the signs).
2. **Cross-Term Audit:** Are all 3 pairwise products ($2ab, 2ac, 2bc$) present?
3. **Peer Signature:** Write one correction note and sign the bottom.

Session Synthesis (10 min)

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FOIL Method	Double distribution rule for multiplying two binomials.

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FOIL Method	Double distribution rule for multiplying two binomials.
Pascal's Triangle	Generates coefficients for $(a+b)^n$ without long expansion.
Variable Powers	a's exponent counts down; b's exponent counts up.
Trinomial Squaring	$(a+b+c)^2 = \sum \text{squares} + 2(\sum \text{pairwise products})$.

Exit Ticket (Google Classroom Upload)

Take a clear photo of your peer-audited notebook page and upload it to **Google Classroom** before leaving class!

50 Minutes of Focused Individual Work

Task Objectives & Instructions

- **Individual Focus:** Work independently on your assigned task/homework.
- **Self-Paced Progression:** Move sequentially through the problem sets or task guidelines.
- **Resource Management:** Consult your reference notes, class slides, or guidelines before asking questions.

Expectations

- Silent and continuous effort.
- Complete all required steps in your notebook/file.

Teacher Support

- Raise your hand for targeted 1-on-1 feedback.
- Check progress against the success criteria.

Timer: 50:00 min – Maximize your active working time!